

# Analytical Derivation of a Supernova Lightcurve

## A One-Zone Radiation Diffusion Model

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### 1 Physical Setup and Motivation

We want to understand, from first principles, how the bolometric luminosity of a Type II supernova evolves with time. Rather than solving the full radiation hydrodynamics equations, we adopt a *one-zone model*: the entire ejecta is treated as a single, spherically symmetric, homogeneous ball of gas. While simple, this model captures the essential physics that sets the shape and timescale of the lightcurve remarkably well.

The physical picture is straightforward. After core collapse, the supernova shock deposits a large amount of energy into the stellar envelope. This energy is initially trapped as radiation because the ejecta is extremely optically thick – photons cannot freely stream out. As the ejecta expands and thins, radiation gradually diffuses outward. The interplay between this diffusion and the expansion is what shapes the lightcurve: the luminosity rises as photons can more easily escape, peaks at a characteristic timescale when the diffusion time approximately equals the expansion time, and then declines.

We denote the total internal (radiation) energy of the ejecta as  $E(t)$ , the total ejecta mass as  $M$ , and the outer radius of the ejecta as  $R(t)$ . The observed luminosity is  $L(t)$ , which is the radiation flux escaping at the outer boundary.

### 2 Energy Conservation in the Expanding Ejecta

The starting point is energy conservation. The rate of change of the total internal energy of the ejecta has three contributions:

1.  **$PdV$  work**: as the gas expands, it does work against its own pressure, converting internal energy into bulk kinetic energy of the expansion.
2. **Radiative diffusion losses**: energy carried away by photons diffusing outward through the ejecta and escaping at the surface.
3. **Energy sources**: radioactive heating (e.g. from  $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$  decay chains) or, more generally in SN II, the shock-deposited energy and any subsequent radioactive input.

Putting these together, the energy equation is:

$$\boxed{\frac{dE}{dt} = -P\frac{dV}{dt} - L_{\text{diff}}(R, t) + \dot{Q}(t)} \quad (1)$$

Here  $P$  is the pressure,  $V$  is the volume,  $L_{\text{diff}}(R, t)$  is the diffusion luminosity evaluated at the outer edge  $r = R(t)$ , and  $\dot{Q}(t)$  is the total source power from radioactive heating or shock energy. The luminosity  $L(t)$  rises steeply at early times when energy input dominates, peaks at  $t = t_{\text{SN}}$ , and then falls off as the ejecta becomes increasingly transparent and the energy reservoir drains. We now evaluate each term in equation (1).

### 3 Homologous Expansion and the Volume Evolution

A key simplifying assumption is *homologous expansion*. In this regime, every fluid element expands at a velocity proportional to its distance from the center,  $v(r) = r/t$ . The outer edge of the ejecta moves at constant velocity  $v$ :

$$R(t) = vt + R_0 \quad (2)$$

where  $R_0$  is the initial radius at the time of explosion (roughly the progenitor stellar radius). It is convenient to define  $t_0 \equiv R_0/v$ , so that:

$$R(t) = v(t + t_0) \quad (3)$$

This substitution is clean because it means the radius is simply  $v$  times a shifted time variable  $t + t_0$ , and all subsequent expressions inherit this structure naturally. At times  $t \gg t_0$ , we recover  $R \approx vt$ , but we retain  $t_0$  for the full generality needed in the integral solution.

The volume of the spherical ejecta is:

$$V(t) = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi v^3(t + t_0)^3 \quad (4)$$

Taking the time derivative:

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3v^3(t + t_0)^2 = 4\pi v^3(t + t_0)^2 = 4\pi R^2 v \quad (5)$$

### 4 Radiation-Dominated Equation of State and the $PdV$ Term

We assume the gas is *radiation dominated*. For a photon gas in thermodynamic equilibrium, the radiation pressure is one-third of the energy density. With  $u = E/V$  being the mean radiation energy density, we have:

$$P = \frac{u}{3} = \frac{E}{3V} \quad (6)$$

Substituting equations (5) and (6) into the  $PdV$  term:

$$P \frac{dV}{dt} = \frac{E}{3V} \cdot 4\pi R^2 v \quad (7)$$

Since  $V = \frac{4}{3}\pi R^3$ , we have  $4\pi R^2/V = 3/R$ , and therefore:

$$P \frac{dV}{dt} = \frac{E}{3} \cdot \frac{3}{R} \cdot v = \frac{Ev}{R} = \frac{E}{t + t_0} \quad (8)$$

where in the last step we used  $R = v(t + t_0)$ . The associated cooling timescale is  $t_{\text{exp}} = R/v = t + t_0$ , the dynamical expansion time.

## 5 The Diffusion Luminosity

### 5.1 Radiative Diffusion Equation

In the diffusion approximation, the radiative energy flux through a spherical shell of radius  $r$  is:

$$L_{\text{diff}}(r, t) = -\frac{4\pi r^2 c}{3\kappa\rho} \frac{\partial u}{\partial r} \quad (9)$$

where  $\kappa$  is the opacity and  $\rho$  is the local density. The minus sign ensures radiation flows from high to low energy density (outward from the hot center).

### 5.2 One-Zone Gradient Approximation

We assume the internal energy is linearly distributed within the ejecta, highest at the center and falling to zero at the outer edge. In the one-zone approximation, we do not resolve the spatial profile. Instead we estimate the gradient at  $r = R$  using the mean energy density  $\bar{u} = E/V$  and the system scale  $R$ :

$$\left. \frac{\partial u}{\partial r} \right|_{r=R} \approx -\frac{E/V}{R} = -\frac{E}{VR} \quad (10)$$

### 5.3 Luminosity at the Outer Boundary

Evaluating equation (9) at  $r = R$  using equation (10):

$$L(t) = -\frac{4\pi R^2 c}{3\kappa\rho} \left( -\frac{E}{VR} \right) = \frac{4\pi RcE}{3\kappa\rho V} \quad (11)$$

Substituting  $V = \frac{4}{3}\pi R^3$ :

$$L(t) = \frac{4\pi RcE}{3\kappa\rho \cdot \frac{4}{3}\pi R^3} = \frac{cE}{\kappa\rho R^2} \quad (12)$$

We now substitute the mean density  $\rho = M/V = 3M/[4\pi v^3(t + t_0)^3]$  and  $R = v(t + t_0)$ :

$$\kappa\rho R^2 = \kappa \cdot \frac{3M}{4\pi v^3(t+t_0)^3} \cdot v^2(t+t_0)^2 = \frac{3\kappa M}{4\pi v} \cdot \frac{1}{t+t_0} \quad (13)$$

It is natural to define the characteristic diffusion timescale of the problem (Numerical factors differ from the classical Arnett solution because of the simplified one-zone treatment adopted here.):

$$t_m \equiv \sqrt{\frac{3\kappa M}{4\pi v c}} \quad (14)$$

so that  $\kappa\rho R^2 = t_m^2 c / (t+t_0)$ , and therefore the diffusion timescale is:

$$t_{\text{diff}}(t) \equiv \frac{\kappa\rho R^2}{c} = \frac{t_m^2}{t+t_0} \quad (15)$$

Notice that  $t_{\text{diff}}$  *decreases* over time as the ejecta expands and thins. The luminosity from equation (12) then takes the compact form:

$$L(t) = \frac{cE}{\kappa\rho R^2} = \frac{E}{t_{\text{diff}}} = \frac{E(t+t_0)}{t_m^2} \quad (16)$$

## 6 The Full Governing ODE

Substituting equations (8) and (16) into the energy equation (1):

$$\frac{dE}{dt} = -\frac{E}{t+t_0} - \frac{E(t+t_0)}{t_m^2} + \dot{Q}(t) \quad (17)$$

Rearranging into standard linear ODE form:

$$\frac{dE}{dt} + E \left( \frac{1}{t+t_0} + \frac{t+t_0}{t_m^2} \right) = \dot{Q}(t) \quad (18)$$

This is a first-order linear ODE of the form  $dE/dt + p(t)E = q(t)$ , with:

$$p(t) = \frac{1}{t+t_0} + \frac{t+t_0}{t_m^2} \quad \text{and} \quad q(t) = \dot{Q}(t) \quad (19)$$

## 7 Solving the ODE via Integrating Factor

### 7.1 Finding the Integrating Factor

For a linear ODE of the form  $dE/dt + p(t)E = q(t)$ , the integrating factor is:

$$\mu(t) = \exp\left(\int p(t) dt\right) \quad (20)$$

We compute the integral of  $p(t)$ :

$$\int p(t) dt = \int \frac{dt}{t+t_0} + \int \frac{t+t_0}{t_m^2} dt = \ln(t+t_0) + \frac{(t+t_0)^2}{2t_m^2} \quad (21)$$

Therefore the integrating factor is:

$$\mu(t) = \exp\left[\ln(t+t_0) + \frac{(t+t_0)^2}{2t_m^2}\right] = (t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) \quad (22)$$

## 7.2 Applying the Integrating Factor

Multiplying both sides of equation (18) by  $\mu(t)$ :

$$(t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) \frac{dE}{dt} + (t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) \cdot p(t) \cdot E = (t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) \dot{Q}(t) \quad (23)$$

By design, the left-hand side is a perfect total derivative:

$$\frac{d}{dt} \left[ (t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) E(t) \right] = (t+t_0) \exp\left(\frac{(t+t_0)^2}{2t_m^2}\right) \dot{Q}(t) \quad (24)$$

We can verify this explicitly. Differentiating the left-hand side using the product rule:

$$\begin{aligned} \frac{d}{dt} \left[ (t+t_0) e^{(t+t_0)^2/2t_m^2} E \right] &= e^{(t+t_0)^2/2t_m^2} E + (t+t_0) \cdot \frac{t+t_0}{t_m^2} e^{(t+t_0)^2/2t_m^2} E + (t+t_0) e^{(t+t_0)^2/2t_m^2} \frac{dE}{dt} \\ &= (t+t_0) e^{(t+t_0)^2/2t_m^2} \left[ \frac{dE}{dt} + E \left( \frac{1}{t+t_0} + \frac{t+t_0}{t_m^2} \right) \right] \\ &= (t+t_0) e^{(t+t_0)^2/2t_m^2} \dot{Q}(t) \end{aligned} \quad (25)$$

confirming that equation (24) is exactly our ODE (18) times  $\mu(t)$ .

## 7.3 Integrating to Find $E(t)$

Integrating both sides of equation (24) from 0 to  $t$ :

$$\left[ (t'+t_0) \exp\left(\frac{(t'+t_0)^2}{2t_m^2}\right) E(t') \right]_0^t = \int_0^t (t'+t_0) \exp\left(\frac{(t'+t_0)^2}{2t_m^2}\right) \dot{Q}(t') dt' \quad (26)$$

which gives:

$$(t + t_0) e^{(t+t_0)^2/2t_m^2} E(t) - t_0 e^{t_0^2/2t_m^2} E_0 = \int_0^t (t' + t_0) e^{(t'+t_0)^2/2t_m^2} \dot{Q}(t') dt' \quad (27)$$

where  $E_0 = E(0)$  is the initial internal energy (set by the shock deposited energy). Solving for  $E(t)$ :

$$E(t) = \frac{e^{-(t+t_0)^2/2t_m^2}}{t + t_0} \left[ t_0 e^{t_0^2/2t_m^2} E_0 + \int_0^t (t' + t_0) e^{(t'+t_0)^2/2t_m^2} \dot{Q}(t') dt' \right] \quad (28)$$

## 8 The Bolometric Lightcurve $L(t)$

The observable quantity is the luminosity. Recalling from equation (16) that  $L(t) = E(t)(t + t_0)/t_m^2$ , we multiply equation (28) by  $(t + t_0)/t_m^2$ :

$$L(t) = \frac{e^{-(t+t_0)^2/2t_m^2}}{t_m^2} \left[ t_0 e^{t_0^2/2t_m^2} E_0 + \int_0^t (t' + t_0) e^{(t'+t_0)^2/2t_m^2} \dot{Q}(t') dt' \right] \quad (29)$$

This is the central result of the model. The lightcurve at any time  $t$  is given by a running integral of the heating history, weighted by an exponential function of  $(t' + t_0)^2$ , and then multiplied by the decaying factor  $e^{-(t+t_0)^2/2t_m^2}$ . Intuitively, this factor represents how efficiently photons deposited at time  $t'$  can diffuse out by time  $t$ : the exponential  $e^{-(t+t_0)^2/2t_m^2}$  is the “diffusion envelope” that suppresses early emission when the ejecta is still thick.

The two terms inside the brackets have clear physical meanings. The first term,  $t_0 e^{t_0^2/2t_m^2} E_0$ , is the contribution from the initial shock-deposited energy  $E_0$ . The second term is the contribution from ongoing heating  $\dot{Q}(t')$ , integrated over all past times weighted by the diffusion kernel. In SN II,  $E_0$  is typically the shock cooling component (dominant at very early times), while  $\dot{Q}$  captures the radioactive nickel/cobalt decay.

## 9 The Peak Timescale

### 9.1 Condition for Photon Escape

The peak of the lightcurve occurs roughly when the diffusion timescale equals the expansion timescale:

$$t_{\text{diff}}(t_{\text{SN}}) = t_{\text{exp}}(t_{\text{SN}}) \iff \frac{t_m^2}{t_{\text{SN}} + t_0} = t_{\text{SN}} + t_0 \iff t_{\text{SN}} + t_0 = t_m \quad (30)$$

so that  $t_{\text{SN}} = t_m - t_0$ . This can be understood physically via the optical depth  $\tau = \kappa \rho R = t_m^2 c / [v(t + t_0)^2]$ . The effective photon diffusion velocity is  $c/\tau$ , and the escape condition is:

$$\frac{c}{\tau} = \frac{c}{\kappa \rho R} \gtrsim v \quad (31)$$

Before the peak,  $c/\tau < v$ : photons are trapped. After the peak,  $c/\tau > v$ : photons escape and the luminosity falls. The transition at  $t = t_{\text{SN}}$  corresponds to  $t + t_0 = t_m$ .

## 9.2 The Peak Timescale

From  $t_{\text{SN}} + t_0 = t_m$  and the definition of  $t_m$ :

$$t_m = \sqrt{\frac{3\kappa M}{4\pi v c}} \quad (32)$$

If the progenitor is compact ( $R_0 \ll vt_m$ , so  $t_0 \ll t_m$ ), then  $t_{\text{SN}} \approx t_m$  and we recover the Arnett scaling:

$$\boxed{t_{\text{SN}} \approx t_m = \sqrt{\frac{3\kappa M}{4\pi v c}} \propto \sqrt{\frac{\kappa M}{v}}} \quad (33)$$

For extended progenitors (relevant to RSG-based SN II),  $t_0 = R_0/v$  is non-negligible and corrects the peak time earlier.

## 10 Application: Heating from Radioactive Nickel

As a concrete application, consider a source term from  $^{56}\text{Ni}$  decay with characteristic timescale  $t_{\text{Ni}}$ :

$$\dot{Q}(t) = Q_0 e^{-t/t_{\text{Ni}}} \quad (34)$$

where  $Q_0$  is the initial heating rate (proportional to the nickel mass  $M_{\text{Ni}}$ ). Substituting into equation (29), the heating integral becomes:

$$\mathcal{I}(t) = \int_0^t (t' + t_0) e^{(t'+t_0)^2/2t_m^2} Q_0 e^{-t'/t_{\text{Ni}}} dt' \quad (35)$$

This integral does not have a simple closed form in general, but it is straightforward to evaluate numerically. The full lightcurve is then:

$$\boxed{L(t) = \frac{Q_0}{t_m^2} e^{-(t+t_0)^2/2t_m^2} \left[ t_0 e^{t_0^2/2t_m^2} \frac{E_0}{Q_0} + \int_0^t (t' + t_0) e^{(t'+t_0)^2/2t_m^2 - t'/t_{\text{Ni}}} dt' \right]} \quad (36)$$

At late times when the ejecta is optically thin ( $t \gg t_m$ ), the luminosity simply tracks the instantaneous radioactive input:  $L(t) \approx \dot{Q}(t) = Q_0 e^{-t/t_{\text{Ni}}}$ . This is Arnett's rule: at peak,  $L_{\text{peak}} \approx \dot{Q}(t_{\text{SN}})$ , directly constraining the nickel mass from the observed peak luminosity.